Super Magnifier LV

User Guide · Version 2026.09.08.0002

Screen magnifier for Windows -- built for low vision users.

Uses the Windows Magnification API for smooth, real-time magnification that works with DirectX games and standard desktop apps.

Current version: **2026.09.08.0002** · Target: **.NET 10** · See `CHANGELOG.md` for release history.

Requirements to Build

Tool	Where
.NET SDK 10 or later	https://dotnet.microsoft.com/download
Windows 10 or 11	--

> The **output exe** targets .NET 10. Publish self-contained to bundle the runtime:

> `dotnet publish -c Release -r win-x64 /p:PublishSingleFile=true /p:SelfContained=true`

> End users need no extra runtime when published self-contained.

How to Build

```
cd SuperMagnifier_build
dotnet build -c Release
```

Output: `bin\Release\net10.0-windows\SuperMagnifierLV.exe`

Features

Magnification

Feature	Details
Zoom range	1.25× to 20×
Zoom step	0.1×, 0.25×, 0.5×, 1×, or 2× per click
Zoom preset buttons	Quick-access 2×, 3×, 4×, 6×, 8×, 12× buttons
Zoom controls	^v buttons, slider, Ctrl+= / Ctrl+-
Smooth animation	Ease-in-out zoom transitions (toggleable in Settings)
Fine zoom mode	Ctrl+G -- temporarily halves the zoom step
Home zoom	Ctrl+0 -- returns to a saved default zoom level
Zoom lock	Clamp zoom between a configurable min and max
Per-monitor zoom	Each display remembers its own zoom level

Lens Window

Feature	Details
Resizable	Drag any edge or corner
Draggable	Drag the title bar
Always on Top	Stays above games and other apps
No focus steal	Never takes keyboard or mouse focus from other apps
Live zoom in title	Title bar shows the current magnification level
Title bar sizes	5 sizes (26-80 px) -- cycle with the Aa button
Lens shapes	Rectangle (fixed), Circle (follows cursor), Square (follows cursor)
Opacity control	Ctrl+[/ Ctrl+] -- fade the lens in or out
Opacity follows zoom	Lens fades automatically at lower zoom levels (optional)
Freeze	Ctrl+F -- freeze the current frame
Click-through	Ctrl+T -- pass mouse clicks to the window below
Snapshot	Save the current lens view as a PNG to the Pictures folder
Mirror lens	Duplicate lens window on a second monitor
Dock to corner	Snap the lens to any screen corner via the Lens panel
Dim when idle	Lens fades after the cursor is still; restores on movement
Keyboard resize	Ctrl+Shift+Arrow -- grow or shrink the lens by 20 px
Off-screen recovery	Lens and cursor-highlighter overlay automatically re-clamp to visible screen area the moment a monitor is added, removed, or its resolution/DPI changes -- not just at startup

Follow Modes

Feature	Details
Follow cursor	Lens tracks the mouse in real time
Follow caret	Lens tracks the text insertion point when typing
Snap-to-Word	Centre the lens on the current word under the cursor
Edge dead zone	Cursor must reach a screen edge before the lens pans
Smooth panning	Lens glides to follow -- adjustable lag in Settings

Colour & Rendering

Feature	Details
Rendering modes	Smooth (bilinear) or Crisp (nearest-neighbour)
Colour filters	Normal, Inverted, Grayscale, High Contrast, Deuteranopia, Protanopia, Tritanopia, Warm Amber, Cool Blue
Colour tint	Adjustable tint intensity for Warm Amber and Cool Blue
Colour cycle	Ctrl+Shift+C -- cycles through all colour filters
Colour picker	Click inside the lens to sample any pixel's hex value
Colour history	Last 10 picked colours saved and shown as swatches

Overlays

Feature	Details
Reading ruler	Horizontal tinted bar that tracks across the screen
Reticle	Cross, X, or circle drawn over the cursor position in the lens
Cursor highlighter	Coloured ring drawn around the cursor
Cursor finder	Ctrl+P -- animated expanding rings to locate the cursor
Focus ring	Ring drawn around the keyboard-focused UI element (colour configurable)

Presets

Feature	Details
5 named presets	Store zoom, lens size, colour filter, opacity, follow mode, snap-to-word, reticle, and zoom lock
Load hotkeys	Ctrl+1 ... Ctrl+5
Save hotkeys	Ctrl+Shift+1 ... Ctrl+Shift+5
Tray quick-load	Right-click tray icon -> Load Preset submenu
Per-app auto-presets	Automatically switch preset when a specific application becomes active
Export / import	Share presets as JSON files independently of other settings

Position Memory

Feature	Details
Per-monitor zoom	Each display remembers its own zoom level
Per-monitor position	Lens restores to the right position on each monitor
Per-app position	Lens position remembered per application
Session restore	Restores zoom, active state, and position from the previous session

Settings

Feature	Details
Settings dialog	Two-column layout with inline validation
Startup defaults	Set default follow mode, snap-to-word, reticle, UI size, and lens title bar size
Export / import	Save and load all settings as a JSON file
Reset to defaults	One-click factory reset in the Settings dialog
Atomic save	Write-to-temp-then-rename prevents corrupt settings on crash or power loss
Settings location	`%AppData%\Roaming\SuperMagnifierLV\magnifier_settings.json`

Hotkeys

Feature	Details
Fully remappable	All hotkeys customisable via the Hotkeys dialog
Remap dialog	Shows all actions with current bindings; preset slots shown read-only
Default bindings	See hotkey table below

Accessibility

Feature	Details
Screen reader support	All controls have accessible names and help text
WCAG 1.4.3	Disabled state tokens; muted text meets 4.5:1 contrast
WCAG 2.1.1	Lens toolbar fully Tab-navigable
WCAG 2.1.2	Escape closes all panels and dialogs -- no keyboard traps
WCAG 2.2.1	Status toasts dismissible by click; extended on reduced-motion systems
WCAG 2.3.3	All animations respect Windows "Show animations in Windows" preference
WCAG 2.4.7	Keyboard focus rings on all interactive controls
WCAG 2.4.11	Focus ring colour adapts to dark/light theme for 3:1 contrast
WCAG 2.5.5	All buttons meet 44×44 px minimum touch target
WCAG 3.3.1	Inline validation errors shown as fields change in Settings
WCAG 4.1.2	Colour swatches are buttons with accessible role and keyboard activation
WCAG 4.1.3	Status toasts announced to screen readers as live regions
High Contrast	Automatically adapts when Windows High Contrast mode is active
Reduce Motion	Skips animations when Windows "Show animations" is off
System font scale	Responds to Windows "Make text bigger" accessibility setting
F1 shortcut	Opens the About dialog from anywhere in the control panel

System Integration

Feature	Details
System tray	Minimize to tray; right-click for quick actions
Autostart	Optional Windows startup registry entry
Single instance	Second launch shows a message instead of opening a second window
Sleep/wake recovery	Automatically reinitialises the magnification engine after system wake
What's New dialog	Shows changelog once per version after an update
First-run wizard	Guided introduction for new users

Project Structure

File	Purpose
App.xaml/.cs	WPF entry point; single-instance mutex; session-ending save
MainWindow.xaml/.cs	Main control panel -- all UI sections and hotkey wiring
MainWindow.Zoom.cs	Zoom logic partial class
MainWindow.Hotkeys.cs	Hotkey registration partial class
LensWindow.xaml/.cs	Floating magnifier overlay; toolbar; colour picker; snapshot
MirrorLensWindow.cs	Second read-only lens window
ReticleOverlayWindow.xaml/.cs	Reticle and cursor-dot overlay
SettingsDialog.xaml/.cs	Settings modal -- two-column layout; inline validation
HotkeyRemapDialog.xaml/.cs	Remap all hotkeys; preset slots shown read-only
AboutDialog.xaml/.cs	Version, settings path, donate links, README button
WhatsNewDialog.xaml/.cs	Changelog shown once per version after update
FirstRunWizard.xaml/.cs	New-user onboarding wizard
MagnifierEngine.cs	Win32 Magnification API wrapper
AppSettings.cs	JSON settings persistence -- hand-rolled parser, no dependencies
HotkeyBinding.cs	Remappable hotkey data, defaults, and display labels
GlobalHotkey.cs	RegisterHotKey / UnregisterHotKey wrapper
CaretTracker.cs	UIAutomation caret position tracking
WordTracker.cs	MSAA word boundary snapping (async, 200 ms timeout)
CursorHighlighter.cs	Coloured ring overlay around the cursor
CursorPulse.cs	Animated pulse effect overlay
ReadingRuler.cs	Horizontal reading guide overlay
LensPreset.cs	Preset load/save and JSON serialisation
AppPresetMap.cs	Per-application automatic preset switching
AppPositionMap.cs	Per-application lens position memory
LensKeyboardMover.cs	Arrow-key lens movement in click-through mode
PerMonitorZoomMap.cs	Per-monitor zoom memory serialisation
MonitorHelper.cs	DPI-aware monitor and position resolution
NativeMethods.cs	All P/Invoke declarations
app.manifest	DPI awareness and UAC level

Settings File

Settings are stored at:

```
%AppData%\Roaming\SuperMagnifierLV\magnifier_settings.json
```

Edit by hand or use the Settings dialog. The file path is shown in the About dialog with a one-click copy button.

Hotkeys (defaults -- all remappable except preset slots)

Action	Default
Toggle magnifier	Ctrl+M
Freeze / unfreeze	Ctrl+F
Zoom in	Ctrl+=
Zoom out	Ctrl+-
Click-through	Ctrl+T
Snap to cursor	Ctrl+Home
Cursor finder pulse	Ctrl+P
Home zoom	Ctrl+0
Fine zoom toggle	Ctrl+G
Opacity down	Ctrl+[
Opacity up	Ctrl+]
Toggle follow mode	Ctrl+L
Toggle reticle	Ctrl+R
Cycle colour filter	Ctrl+Shift+C
Load preset 1-5	Ctrl+1 ... Ctrl+5
Save preset 1-5	Ctrl+Shift+1 ... Ctrl+Shift+5
Resize lens up	Ctrl+Shift+Up
Resize lens down	Ctrl+Shift+Down
Resize lens left	Ctrl+Shift+Left
Resize lens right	Ctrl+Shift+Right

Tips for Games

1. Run your game in **Borderless Windowed** mode.
2. Enable **Always on Top** in Super Magnifier LV.
3. Press **Ctrl+M** -- the lens will appear.
4. Drag the lens to position; resize by dragging any edge or corner.
5. The lens never steals keyboard or mouse focus from your game.

Notes

- **YoloMouse / custom cursor tools:** The lens window intercepts `WM_SETCURSOR` to hide the system cursor inside the lens. If you use a cursor-replacement tool, uncheck **Hide cursor inside lens** in Settings -> Zoom & Animation.
- **Colour filter + tint:** Tint intensity only applies to Warm Amber and Cool Blue modes.
- **Snap-to-Word** uses MSAA on a background thread with a 200 ms timeout so it never blocks the UI.

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